INDIVIDUAL REPORT

Pacific BioArchive: Multi-Cloud Serverless Data Platform

Student	Caoyu Jiang (蒋曹羽)
Student ID / Monash ID	cjia0077
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1. Role and Contribution

My primary role was authentication and AWS cloud infrastructure. I translated the security requirements into a reproducible AWS SAM design centred on Amazon Cognito, API Gateway JWT authorisation, private S3 buckets, Lambda functions, DynamoDB and SNS. On the user path, I contributed the custom registration, email confirmation, sign-in, sign-out and protected-route flow, including first name and last name attributes and the Google federation callback. On the service path, I connected Cognito access tokens to protected API routes and documented how the same token is verified by the secondary cloud. I also worked on deployment and configuration scripts so a temporary AWS Academy environment could be rebuilt consistently without committing credentials. My contribution focused on least-privilege boundaries, private storage, pre-signed uploads and deploy-time configuration rather than browser-held cloud keys. I validated the frontend production build and reviewed the infrastructure template against the assignment rubric, especially unauthenticated access, cross-cloud token handling and IAM permissions.

2. Teamwork Reflection

Working with three teammates made the system dependencies much clearer than an isolated implementation would have. Ze's upload and ML-processing work defined the events, object keys and processing states that my infrastructure had to support. Yang Chenjun's data-management and notification work depended on consistent ownership claims and permissions, while Liang Xuehao's Alibaba Cloud query path exposed whether our Cognito token contract was truly portable across providers. The main challenges were coordinating interfaces while AWS Academy sessions expired, handling a large model-based Lambda image, and resolving CORS and URL-shape differences between AWS and Alibaba Cloud. We worked most effectively when we agreed on shared schemas first, recorded deployment steps, and added focused tests around each boundary. Integration also showed that a locally correct component can still fail when permissions, event triggers or runtime configuration are incomplete. I learned to communicate assumptions early, keep infrastructure reproducible, and treat tests and documentation as team interfaces rather than final-stage tasks.

3. Generative AI Statement

Generative AI was used selectively in this assessment. I used OpenAI Codex to help review the repository against the published requirements, organise draft wording, check build and test results, and verify document formatting. I reviewed and revised the generated suggestions, checked technical claims against the implemented source code and assignment documents, and remain responsible for understanding, explaining and modifying the submitted work. No credentials or generated claims are included in this report.

Word count: 393 (excluding title, headings and metadata)